CLAUDE CODE TELEMETRY ANALYTICS PLATFORM

Internship Technical Assignment

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PROJECT GOAL

The goal of this project was to build an analytics platform that:

- processes raw telemetry event streams
- stores structured data in a database
- extracts meaningful usage insights
- visualizes patterns through an interactive dashboard

Technologies used
Python
SQLite
Pandas
Streamlit
Matplotlib

SYSTEM ARCHITECTURE

Telemetry Logs (JSONL)

↓

ETL Pipeline (Python)

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SQLite Database

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Analytics Queries

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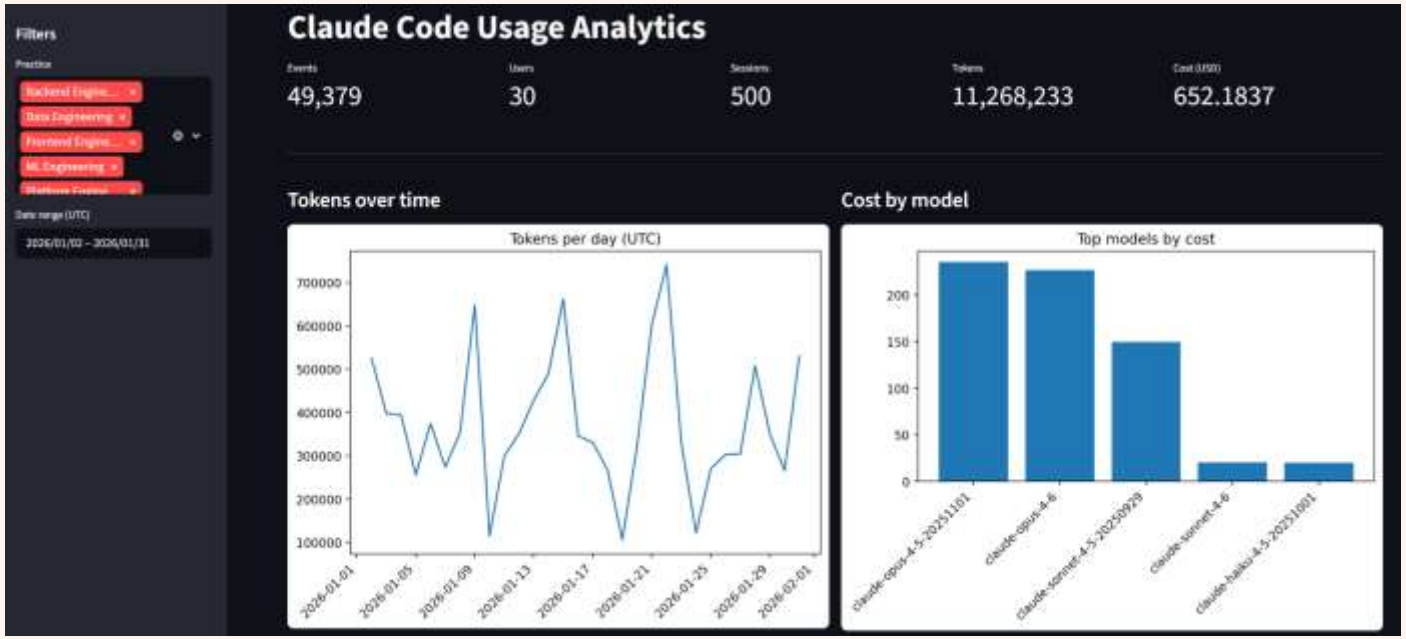
Streamlit Dashboard

The pipeline transforms raw telemetry events into structured analytics insights.

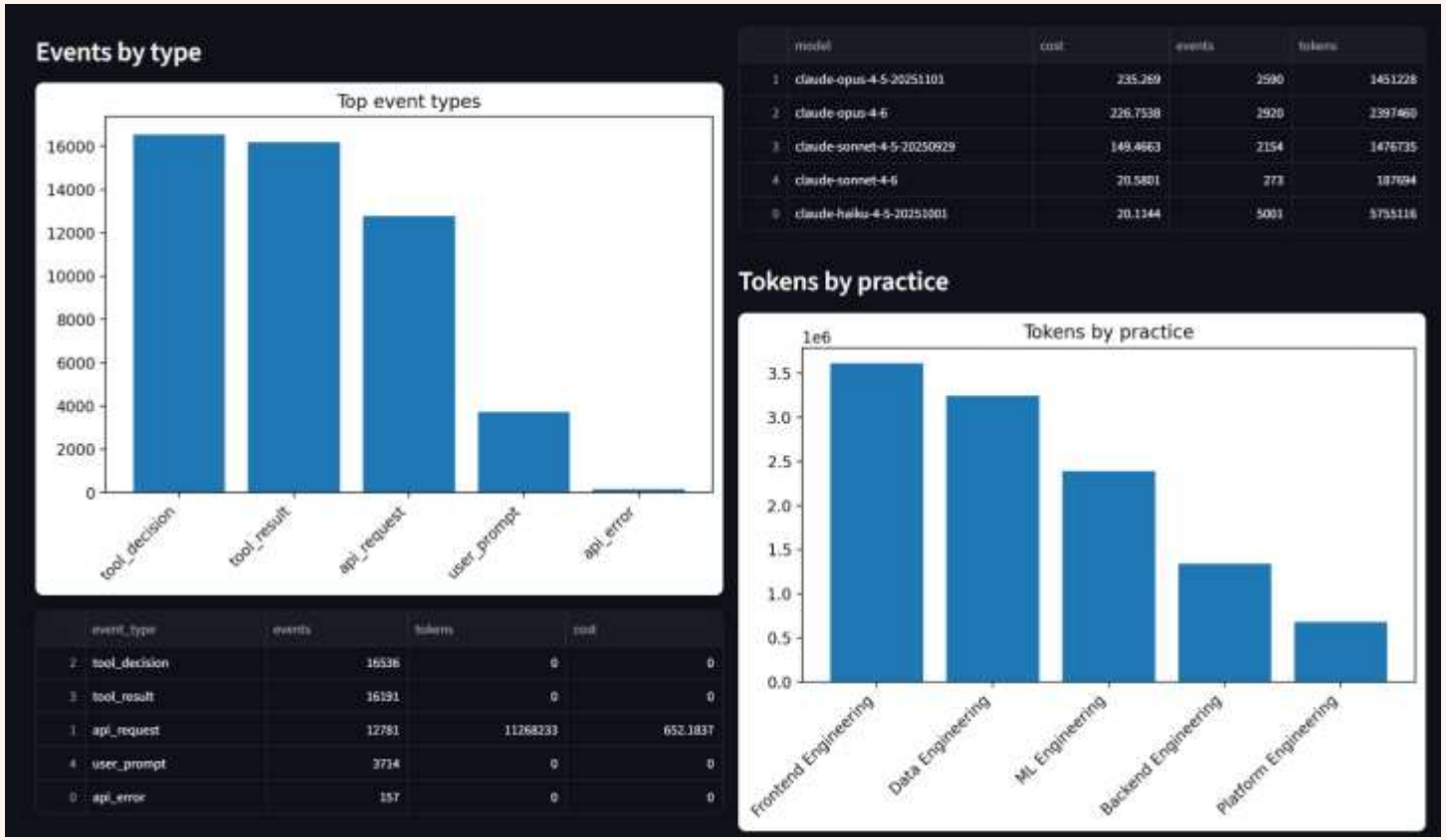
DASHBOARD OVERVIEW



Key metrics from the dataset:

- 49,379 telemetry events processed
- 30 developers
- 500 sessions
- 11.2M tokens consumed
- ~$652 total model cost

KEY OBSERVATIONS FROM THE TELEMETRY DATA



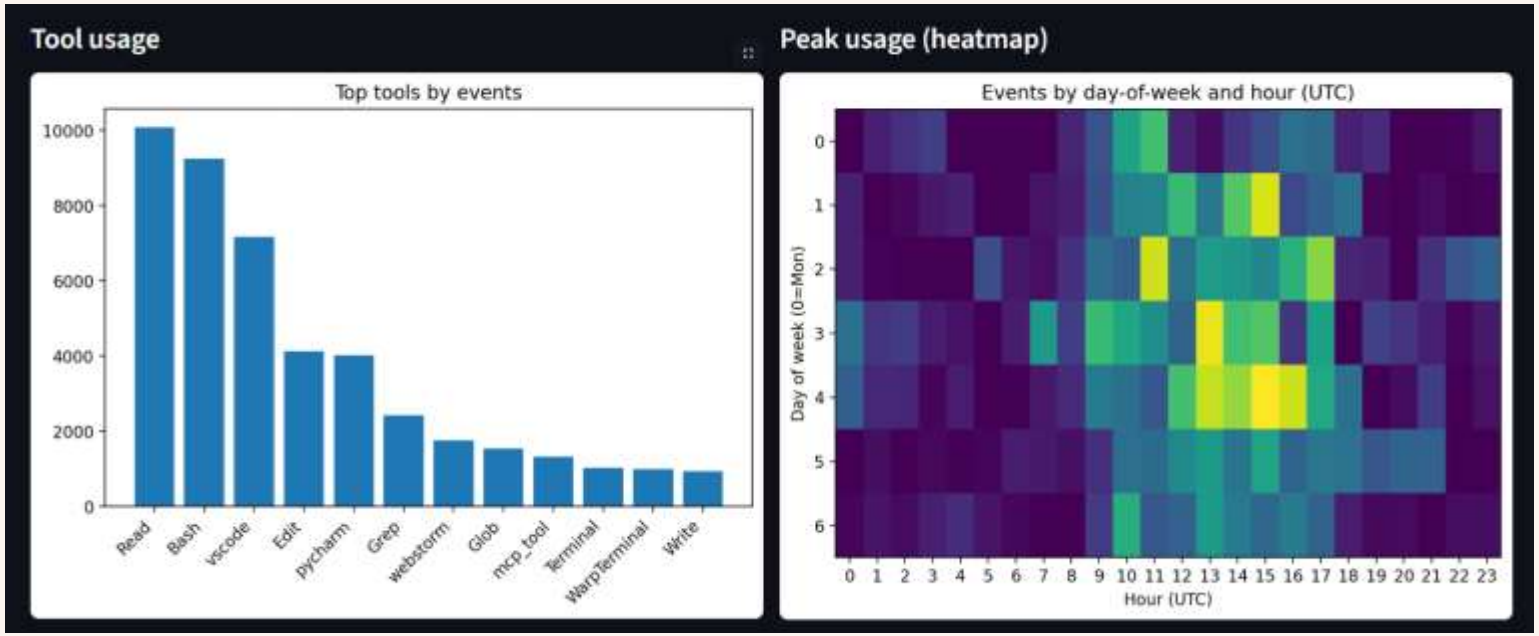
	model	cost	events	tokens
1	claude-opus-4-5-20251101	235.269	2590	1451228
2	claude-opus-4-6	226.7538	2920	2397460
3	claude-sonnet-4-5-20250929	149.4663	2154	1476735
4	claude-sonnet-4-6	20.5801	273	187694
0	claude-haiku-4-5-20251001	20.1344	5001	5755116

	event_type	events	tokens	cost
2	tool_decision	16536	0	0
3	tool_result	16191	0	0
1	api_request	12781	11268233	652.1837
4	user_prompt	3714	0	0
0	api_error	157	0	0

- Tool-related events dominate developer workflows
- tool_decision and tool_result are the most common event types
- API requests generate the majority of token usage and operational cost
- Frontend and Data Engineering practices consume the most tokens

DEVELOPER TOOL USAGE

	tool	events	tokens
7	Read	10070	0
1	Bash	9233	0
23	vscode	7160	4774094
2	Edit	4127	0
22	pycharm	4018	2751678
5	Grep	2416	0
24	webstorm	1755	1188011
4	Glob	1527	0
21	mcp_tool	1323	0
11	Terminal	1016	722760



Developers heavily rely on tools while interacting with Claude Code.

Most commonly used tools:

- Read
- Bash
- VSCode
- Edit

This indicates that Claude Code often acts as an orchestrator between developer tools.

CONCLUSION

The platform successfully transforms raw telemetry data into actionable insights through:

- automated data ingestion
- structured storage
- analytics queries
- interactive visualizations

Future improvements

- real-time telemetry ingestion
- anomaly detection
- model cost optimization